



CONFINED SPACE PROGRAM

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TriCounty Metropolitan Transportation District of Oregon (TriMet)

Safety & Security Division

1020 NE 1st Ave, Portland, OR 97232



1. Policy

This program has been developed to ensure the safety of personnel working in confined spaces.

This program is part of TriMet's environmental, health, and safety plan and serves to meet Federal OSHA and Oregon OSHA Incident Prevention Plan (IPP) and Accident Prevention Plan (APP) requirements.

TriMet's policy is to provide employees with a safe and healthy work environment, and compliance with the procedures and practices described in this program is mandatory to achieve this goal.

2. Authority

This document is administered by the Safety Department in compliance with TriMet's Agency Safety Plan, SOP 230, and includes, but is not limited to, the following regulatory rules:

- OAR Chapter 437, Division 2, Rule 437 437-002-0146
- OSHA 1910.146 Permit-Required Confined Spaces

3. Scope

The policy is applicable to all TriMet employees, contract and temporary workers, and all TriMet facilities and operating locations who enter confined spaces. Employees and their representatives access this Confined Space written program through the TriNet Safety page.

4. Positions and Responsibilities

POSITIONS	RESPONSIBILITIES
Senior Management	• Actively support this program by ensuring its contents are followed and upheld by all employees. • Provide leadership, resources, and program support. • Provide funding commensurate with this program's requirements. • Ensure individual accountability for compliance with this program. • Ensure an environment where employees are encouraged to follow these policies and procedures.
Program Administrator Safety Management Systems Manager	• Review the effectiveness of the TriMet Confined Space Entry Program and update annually. • Ensure compliance with standards set forth in the program by periodic. • Inspection of entry sites and canceling permits where unsafe conditions are present, or where safe procedures are not in practice.



	• Perform an annual review of all permit required confined space entries to ensure compliance with the program. • Assist managers and supervisors with: • Providing training. • Identifying and evaluating permit required confined spaces. • Permit required confined spaces labeling requirements.
Managers, Assistant Managers and Supervisors	• Set the example and consistently ensure full implementation of the TriMet Confined Space Program for TriMet employees, contractors, vendors, and visitors. • Be trained as a Confined Space Supervisor per OAR 437-002-0146. • Ensure that all employees assigned to work in confined spaces are adequately trained and authorized to perform confined space work, and that training records are maintained. • Ensure a Job Hazard Analysis is conducted for Confined Space tasks assigned under their supervisor. • Develop a list of identified confined spaces with unique identifiers at each location. • Review and audit all Confined Space Permits and ensure that cancelled permits are turned into the Safety Department. • Provide employees with required appropriate PPE when required and ensure it properly fits each employee that it is assigned to. • Perform periodic inspections of work areas and tasks under their supervision to identify areas and tasks in which appropriate PPE is required and being used properly. • Ensure systems are in place to verify visitors and contractors comply with Confined Space equipment requirements found in this program. • Ensure confined space equipment that is non-compliant, damaged, or that has been altered from the manufacturer's original specifications is removed from service immediately. • In the case of an incident or a near-miss, inspect confined space equipment for damage, and replace it when non-compliant, damaged, or if it has been altered from the manufacturer's original specification. • Ensure employees are properly inspecting and storing their confined space equipment. • Each department shall assign a Competent Person responsible for departmental confined space equipment program administration and maintenance including equipment inspections, inventory, and recordkeeping.
Project Managers	• Be trained as a Confined Space Supervisor per OAR 437-002-0146. • Include in bid specifications or work plans the expectations for compliance with confined space procedures. • Review the contractor's written permit required confined space program for compliance with OSHA regulations, and the TriMet Confined Space Program.

	• The contractor's program must meet or exceed the requirements found in the OSHA standard, and any additional requirements set forth in the TriMet program. • Inform contractors of work in confined spaces and of the specific actual or potential confined space hazards. • Notify the Safety Department when a contractor confined space entry will occur by emailing: confinedspace@trimet.org. • Periodically audit the contractor's fieldwork. NOTE: Failure to comply with this plan could result in work being discontinued. • If any hot work is performed inside the confined space, ensure that the TriMet hot work permit is completed and followed. • Discuss entry operations with the employer after they are completed. This discussion must include: The program followed during permit space entry and any hazards confronted or created. • When employees enter a permit required confined space under the control of another entity, at the conclusion of entry operations, they must inform the controlling contractor and host employer about the precautions and procedures followed and any hazards that were present or that developed during entry operations.
Responsibilities for Contractors	• In the case of multiple contractors, the Controlling (General) Contractor is responsible for coordinating entries of all contractors and subcontractors. • In addition to complying with OAR confined space requirements, contractors shall: • Contact the TriMet project manager and review confined space requirements. The project manager is encouraged to consult with the Safety department for support as needed. • Submit a copy of each company's permit required confined space program, and training certifications to the project manager for review and approval. • Submit a confined space entry work plan for approval by the project manager. • Provide all confined space equipment and personnel necessary to comply with Safety standards.
Confined Space (CS) Entry Supervisors	• Be trained as a Confined Space Supervisor per OAR 437-002-0146. • Identify and evaluate spaces for classification as permit required or alternate entry. • Review, approve, and sign or reject the confined space permit prior to the entry. • Be familiar with the signs, symptoms and consequences of exposures to chemicals when exposure can reasonably be anticipated. • Evaluate confined space hazards and ensure proper controls are in place. • Determine the responsibility for permit requirements and hazard controls.

	• Determine, whenever responsibility for a permit space entry operation is transferred, that entry operations remain consistent with terms of the entry permit, and that acceptable entry conditions are maintained. • Verify rescue procedures are in place. • Obtain and verify the functionality of all equipment necessary to complete the confined space entry project. • Authorize entry by signing the Entry Authorization Space on the entry permit after verifying all conditions for a safe entry have been met. • Post the permit near the space. • Ensure that the required atmospheric tests are performed in the confined space and record results on the permit prior to entry authorization. • Terminate the entry and cancel the permit when: • Entry operations covered by the entry permit have been completed. • A condition that is not allowed under the entry permit arises in or near the permit space. • The correct procedures for safely entering the space are not followed. • Remove unauthorized individuals who enter or who attempt to enter the permit space during entry operations. • Send a copy of the canceled permit to the Site Manager and the Safety Department (confinedspace@trimet.org).
Attendant	• Know the hazards they may be facing during entry, including information on the type of hazard, as well as signs, symptoms, and consequences of exposure to those hazards. • Be aware of possible behavioral effects of hazard exposure in authorized entrants. • Continuously maintain an accurate count of authorized entrants in the permit space and ensure that the means used to identify authorized entrants accurately identifies who is in the permit space. • Remain outside the permit space during entry operations until relieved by another attendant. • Communicate with authorized entrants as necessary to monitor entrant status and to alert entrants of the need to evacuate the space. • Monitor activities inside and outside the space to determine if it is safe for entrants to remain in the space and order the authorized entrants to evacuate the permit space immediately under any of the following conditions: • If the attendant detects a dangerous or hazardous condition. • If the attendant detects the behavioral effects of hazard exposure in an authorized entrant. • If the attendant detects a situation outside the space that could endanger the authorized entrants or • If the attendant cannot effectively and safely perform all the duties required of the attendant.

	• Summon rescue and other emergency services as soon as the attendant determines that authorized entrants may need assistance to escape from permit space hazards. • Take the following actions when unauthorized persons approach or enter a permit space while entry is underway: • Warn the unauthorized persons that they must stay away from the permit space. • Advise the unauthorized persons that they must exit immediately if they have entered the permit space and • Inform the authorized entrants and the entry supervisor if unauthorized persons have entered the permit space. • NOTE: The employer can give the attendant the authority to remove unauthorized individuals who enter or who attempt to enter the permit space during entry operations, as long as the attendant does not enter the space. • Perform non-entry rescues as specified by the employer's rescue procedure and • Perform no duties that might interfere with the attendant's primary duty to monitor and protect any authorized entrant.
Entrant	• Know the hazards they may be facing during entry, including information on the type of hazard, as well as signs, symptoms, and consequences of exposure to those hazards. • Communicate with the attendant as necessary so the attendant can monitor the entrant's status and to enable the attendant to alert entrants of the need to evacuate the space. • Alert the attendant whenever the entrant detects a dangerous or hazardous condition or warning sign or symptom of exposure to a dangerous situation. • Exit from the permit space as quickly as possible whenever: • An order to evacuate is given by the attendant or the entry supervisor. • The entrant recognizes any warning sign or symptom of exposure to a dangerous situation. • The entrant detects a dangerous or hazardous condition, or an evacuation alarm is activated.
Rescue Team	• Evaluate spaces to make sure their rescue team has the skills and equipment available to perform an entry rescue for spaces they are assigned to. • Ensure rescue team members are working in an area that allows them to have an appropriate response time. • Inform the confined space supervisor if the team becomes unavailable.

5. Confined Space Procedures

5.1. Identifying Confined Spaces

Employees must be able to recognize a confined space. A Confined Space has all the following characteristics:

- It's large enough that an employee can enter and perform work.
- It has limited or restricted means for entry or exit.
- It is not designed for continuous employee occupancy.

Confined space openings are mainly limited by size and location. They may be small and difficult to move through, but in some types, like pits or excavations, they may be very large. Accessing the work area by a fixed ladder constitutes limited or restricted entry or exit. Size and location may also make rescue efforts difficult.

Confined spaces are not designed for employees to enter and work on a routine basis, so they typically will not have good ventilation or lighting. Because of this, the occasional entry for inspection, maintenance, repair, or other tasks may be difficult and hazardous.

All Confined Spaces must be identified at TriMet Facilities or areas where TriMet employees regularly enter or are exposed to Confined Spaces by the manager or designee.

Confined Spaces must be noted by the manager on SAF-8255-FC-CSLog and sent to the confinedspace@trimet.org email.

- *Exception: The locations of permit spaces at remote unmanned locations do not need to be added to the program until the first time employees go to that location after the effective date of this rule.*

5.2. Signs and Labeling to Prevent Unauthorized Entry

Employees must be informed about the danger and location of permit spaces through signs, labels, tags, or equally effective means. An example of sign wording is listed below:

**“Danger, Permit Required Confined Space,
Authorized Personnel Only”.**

All necessary measures must be taken to prevent unauthorized employees from entering permit spaces. Additional measures may be implemented to prevent unauthorized employees from entering permit spaces, such as locks and special tools to gain access.

Employees are prohibited from entering any unevaluated confined spaces until they are fully evaluated.

TriMet employees who are mobile must be able to identify confined spaces through training and the definition of a confined space.



5.3. Evaluating Confined Spaces

All TriMet confined spaces are considered to be Permit Required entries until the Confined Space Entry Supervisor has evaluated the space and determined that the space may be entered as an Alternate Entry. The rationale for classifying confined spaces to Alternate Entry must be detailed on the SAF-8251-CSEvalPermit Page 1 by a trained Confined Space Entry Supervisor.

The Confined Space Evaluation includes initial air monitoring with a bump tested, calibrated multi-gas air monitor, evaluating any known or anticipated hazard, and determining if the space has a characteristic that would make it Permit Required or Alternate Entry. Previous determination, precautions, and procedures for entering the space may be used as part of the evaluation.

Mobile worker employers must evaluate the confined spaces they are responsible for or where their employees will be working during that shift.

On sites where confined spaces are being built, TriMet or the Controlling Contractor is responsible for ensuring this evaluation is accomplished only when:

- Any TriMet employees enter a confined space.
- An agent of TriMet enters a confined space.
- Employees of an employer accountable to the Controlling Contractor or TriMet enter a confined space.
- TriMet or Controlling Contractor assumes control over a confined space.
- All employees and their representatives may observe any part of the evaluation process.

5.4. Hazardous Atmosphere Air Monitoring

Air Monitoring must be done by a trained Confined Space Supervisor or Attendant.

The initial step in evaluating a confined space includes air monitoring, using at least a four gas air monitor that has been bump tested and calibrated before its use.

Calibration gas must meet the National Institute of Standards and Testing (NIST) standards and must not be used if it's outside the vendor expiration date stamped on the label.

Before an employee enters the confined space, test the internal atmosphere with a calibrated, direct-reading instrument for at least the following, in this order:

- Oxygen content: Oxygen content must be between 19.5-23.5%
- LEL Flammable gases and vapors must be 10% or less.
 - *NOTE: Any level of flammable gas is extremely dangerous and must be ventilated down to a 0 level before entering the space.*



- Hydrogen Sulfide H₂S - The ACGIH TLV for hydrogen sulfide is 20 ppm.
- Carbon Monoxide CO - The ACGIH TLV for carbon monoxide (CO) is 25 ppm.
- All atmospheric hazards (except for oxygen) shall be ventilated to zero before entering the space.

Air monitoring must be continuous in permit required confined spaces. If the air hose can remain in the Entrant's breathing zone, the monitor may be transferred to the entrant or left with the attendant.

Air monitoring must be recorded on the Confined Space Evaluation, and the tester's initials must be listed under each documentation time.

5.5. Permit Required Confined Spaces

Permit required confined spaces meet the definition of a confined space and have one or more of the following conditions found during the evaluation completed immediately before entering the space:

- It contains or has the potential to contain a hazardous atmosphere.
- A physical hazard such as:
 - It contains a material that has the potential to engulf an entrant.
 - It has an internal configuration such that an entrant could be trapped or asphyxiated by inwardly converging walls or by a floor that slopes and tapers to a smaller cross-section.
- Contains any other recognized serious safety or health hazard. Examples of serious safety or health hazards include:
 - Chemical hazards
 - Fall, slip, or tripping hazards
 - Electrical hazards
 - Moving machinery parts
 - Extreme ergonomic conditions
 - Hazardous noise levels
 - Biological hazards (mold, fungus, bacteria, sewage, decaying organic matter, etc.)
 - Traffic or track exposure

Examples of permit required confined spaces at TriMet include sewers, manholes, trenches, above and underground storage tanks, sand silos, sumps, electrical vaults, HVAC units with unguarded fans, and hydraulic lift pits.



A permit is required for confined spaces, and a completed permit, continuous air monitoring and ventilation, an authorized attendant, and a documented rescue plan are needed.

5.6. Permit Required Confined Space Entry Procedure

The permit will provide a guide for the Confined Space Entry Supervisor in planning the entry, verifying the rescue plan, documenting acceptable entry conditions, and keeping accountability of the entrants.

The Confined Space Hazard Evaluation lists methods to eliminate and control hazards. A Confined Space Entry Supervisor may add non-listed hazards to the Permit.

The following steps are to be followed by the Confined Entry Supervisor to prepare the confined space for entry (these steps are achieved by completing SAF-8251-PER):

- Ensure that the Confined Space Supervisor, Entrants, and Attendants have current confined space training.
- The entry Supervisor completes the SAF-8251-PER-CSEvalPermit (pages 1-2) and determines the space is Permit Required.
- Determine the rescue plan as Non-Entry or Entry Rescue.
- Contact the Rescue team leader, discuss the rescue plan, and confirm they are available for rescue in the appropriate timeframe in writing.
- Place warning signs or barriers around the confined space to prevent unauthorized entry as necessary.
- Verify traffic control or Method of Track Safety, if required.
- Complete documentation of initial air monitoring.
- Utilize the blower for ongoing ventilation as required. While it may not always be essential for managing atmospheric hazards, it remains a recommended safety practice.
 - *NOTE: Ensure that the gas-powered blower motor is downwind of the confined space.*
- Engulfment lines are disconnected or LOTO.
- Hazardous energy is LOTO, or appropriate SOP is followed.
- SDS for all chemicals have been reviewed, and appropriate controls have been put in place.
- Emergency equipment is inspected and set up.
 - *NOTE: Retrieval devices (tripod, davit, winch, or harness) must be used on all permit entries unless they increase the hazard level or would not contribute to the entrant's rescue.*
- Respiratory equipment is inspected and set up if required.



- Hot Work Permit is completed if any work producing sparks or flames is performed in the space.
- Communication between the Attendant and Entrant is confirmed. Communication between the Attendant and Rescue team is confirmed.
- Entrant wears the multi-gas meter on their belt or lapel throughout the job inside the space. Alternatively, the confined space Attendant may provide continuous air monitoring from outside the space. Monitoring must represent all exposures to entrants.
- The entrant dons a full body harness with a retrieval line secured to a retrieval device. A harness is required if an entry rescue team is used, but not a connected retrieval.
- Attendant(s) and Entrant(s) are noted on the permit.
- The entry Supervisor signs the permit and authorizes entry to begin.
- The entry Supervisor notifies the Supervisor or Manager that entry is commencing.
- If a Track Permit has been issued, the Entry Supervisor notifies TriMet Operations Control Center (OCC) that entry is commencing.
- When work is completed, the permit is updated with exit times. All openings are closed, and systems to prevent unauthorized entry are installed.
- Entry Supervisor cancels the permit and notifies OCC (if notified of entry) once the work has been completed.
- The entry Supervisor forwards the canceled Permit to the site Manager for final review and audit.
- The manager sends the completed Permit to the Safety Department at confinedspace@trimet.org.

5.7. Alternate Entry

A confined space evaluation is required by a confined space entry supervisor on every confined space. Alternate Entry into confined spaces does not require completing the permit, an attendant, or a rescue plan.

Permit spaces may be entered using the Alternate Entry procedure when:

- All hazards have been eliminated, or
- All physical hazards, if any, have been eliminated, and all atmospheric hazards are controlled with continuous ventilation.
 - *Note: For purposes of this rule, tagout alone does not eliminate a hazard. To enter a confined space using alternate entry procedures, a full LOTO system must be used.*



Alternate Entry cannot be used in a continuous system unless conditions that may cause a hazard, such as a hazardous atmosphere or an engulfment hazard, have been eliminated.

The following steps must be followed to enter a confined space using Alternate Entry:

- A Confined Space Entry Supervisor sets up continuous air monitoring, determines the ventilation that must be used, and completes the SAF-8251-PER-CSEvalPermit (Page 1), eliminating all physical hazards.
 - *NOTE: Alternate entry is allowed when fall hazards (if any) have been addressed, all other physical hazards (if any) have been eliminated, and all atmospheric hazards have been eliminated or controlled with continuous ventilation.*
- For an employee to enter the space, air monitoring must read a zero hazard atmosphere.
- If ventilation controls a hazardous atmosphere, Entrants must have time to exit the space safely in the event of a ventilation failure.
- Provide all employees who conduct alternate entry an effective means of communication, such as a two-way radio, cell phone or voice if other employees are present, to summon help while within the space.
- Alternate entry requires another employee working in the area who is in contact with the Entrant (aka "Buddy System").

5.8. Confined Space Evacuation

Entrants must exit the space immediately if a direct reading instrument fails, the ventilation system fails, a hazardous atmosphere is detected, a new hazard is introduced, a hazard develops, an entrant shows signs of exposure, or conditions change in the space.

When a space is evacuated, it cannot be re-entered as an Alternate Entry until the Confined Space Evaluation is updated and the source of the evacuation hazard has been eliminated. If the hazard can only be controlled and not eliminated, the new entry must be permit-required.

NOTE: An attendant may monitor more than one space at a time, but the duties related to one space may not interfere with the duties for any other spaces. If an attendant's attention is focused on one space, such as initiating the rescue procedures, all other spaces that the attendant is monitoring must be evacuated, or another attendant must take over those duties first.

5.9. Rescue Procedures

When multiple entrants are in a permit space, the rescue plan must address how all entrants will be removed in a timely manner.



Where feasible, use non-entry retrieval systems or methods whenever an authorized entrant enters a permit space unless it would increase the entrant's overall risk or would not contribute to the entrant's rescue.

Each authorized entrant must use a chest or full body harness, with a retrieval line attached at:

- The center of the entrant's back near shoulder level
- Above the entrant's head, or
- At another point which, you can establish that presents a profile small enough for the successful removal of the entrant.

Wristlets, ankle straps, or other equally effective means may be used in place of the chest or full body harness if you can demonstrate that the use of a chest or full body harness is infeasible or creates a greater hazard and that the use of other methods are the safest and most effective alternative.

A mechanical device must be set up to retrieve personnel from vertical type permit spaces more than 5' (1.52 m) deep.

All entry rescue teams must be contacted before entry, agree in writing to perform the rescue duties, be certified for Entry Rescue, be capable of performing all rescue operations, and be certified in First Aid and CPR.

Entry Rescue may be used where:

- Non-entry rescue is not feasible or would increase the overall risk to the entrant.
- A trained rescue team with appropriate equipment is available within a reasonable response time.
- The Safety Department has been notified before entry that an Entry Rescue team is being used to meet the rescue requirements.

In the event of an emergency, the attendant should:

- Call TriMet Operations Control Center (OCC). OCC will notify 911, Portland Fire and Rescue, or another emergency team will be deployed, and the TriMet rescue team.
- Initiate the rescue plan.
- If non-entry rescue, the attendant shall attempt to remove the victim by use of the retrieval line from outside the confined space.
- If entry rescue, the attendant contacts the rescue team leader and proceeds with the rescue plan.
- Once the entrant is rescued, rescue team members should administer first aid within the limits of their training until emergency medical services (EMS) arrive.
- The Attendant will provide EMS personnel with any requested information.



If an injured entrant is exposed to a substance for which a Safety Data Sheet (SDS) or other similar written information is required to be kept at the worksite, that SDS or written information must be made available to the medical facility treating the exposed entrant.

5.10. Multi-Employer Work Sites

Project managers must inform all employers and their employees entering TriMet worksites that contain confined spaces that the workplace contains permit spaces and can be entered only when the applicable requirements of this rule are met.

If contractors are entering TriMet-controlled confined spaces, the project manager must inform the contractor of the identified hazards and TriMet's experience with each permit space they will enter.

All contractor entry operations must be coordinated with the project manager when employees of different employers will be working in or near the same permit spaces.

After contractor entry operations are complete, contractors must have a post entry meeting that reviews the program and permit processes followed during permit confined space entry and any hazards confronted, modified, or created.

When TriMet employees enter a permit space under the control of another entity, at the conclusion of entry operations, the manager or designee will provide a copy of the completed Confined Space Evaluation and Permit to inform the controlling contractor or host employer about the precautions and procedures TriMet workers followed and any hazards present or developed during entry operations.

6. Training

No employee shall work in any of the confined space roles or enter a confined space unless they have been properly trained in confined space entry procedures and this program plan per the OAR 437-002-0146(11) requirements.

TriMet provides training to each employee involved in permit required confined space activities, so they acquire the understanding, knowledge, and skills necessary to perform assigned duties safely. Training must establish employee proficiency in stated confined space duties as required by

The TriMet Safety Department will audit, amend and authorize all confined space related training.

The training must be in a language and vocabulary employees understand so they acquire the understanding, knowledge, and skills necessary to perform assigned duties safely.

Observing employee performance during training exercises that simulate actual confined space conditions, as well as a comprehensive written examination, will establish proficiency for each confined space role.



For non-entry and entry rescue training, training sessions will be held for permit required confined space rescues at least once **every 12 months** where dummies, manikins, or actual persons are removed from either:

- The actual permit spaces, or
- Representative permit spaces that simulate the opening size, configuration, and accessibility, of permit spaces where rescue will be performed.

Annual rescue training will be used to introduce new or revised procedures as necessary.

6.1. Training Documentation

Ensure the documentation:

- Contains the employee's name, name and signature of the trainer, training date, and certification expiration date.
- Contains the responsibilities for which they were trained.
- Is available for inspection by employees and their authorized representatives.



6.2. Training Guides

Training will be assigned to each group as follows:

POSITIONS	GUIDE NUMBER	TRAINING TOPIC	FREQUENCY
Manager, Supervisor, Assistant Supervisor and Qualified Protection Persons	SAF-8255-TG-SAER	Confined Space Supervisor, Attendant, Entrant and Non-IDLH Rescue	Every 3 years
All employees exposed to Confined Spaces in the course of their work	SAF-8254-TG-CSA	Confined Space Awareness Training Guide <i>Note: Does not allow you to enter confined spaces.</i> <i>Note: Awareness training is not required for employees whose exposure is negligible.</i> <i>Similarly, when all permit spaces cannot be accessed or opened by employees, awareness training is not required.</i>	Every 3 years

All employees who enter Confined Spaces	SAF-8255-TG-SAER	Confined Space Supervisor, Attendant, Entrant and Non-IDLH Rescue	Every 3 years
All employees who enter Confined Spaces	SAF-8256-TG-SAER-R	Confined Space Supervisor, Attendant, Entrant and Non-IDLH Entry Refresher Training Guide	Annual

7. Equipment and Inspection Requirements

The SMS Manager or designee is responsible for selecting all confined space equipment.

The following equipment will be provided, used, and maintained per manufacturer's recommendations.

Managers and Supervisors are responsible for maintaining all confined space equipment.



EQUIPMENT	USE	INSPECTION FREQUENCY
Barriers or shields, such as pedestrian, vehicle or other barriers	Protecting employees from hazards outside of the space and preventing unauthorized persons from entering the space.	Before every use.
Chemical Protection Suit (CPC)	Protective clothing that shields the wearer from hazardous chemicals, toxins and other dangerous substances. Also known as hazmat suits.	Before and after every use.
Communication equipment	Effective communication between the attendant and the entrants and to initiate rescue when required.	Batteries charged on radios. Communication checks before entry.
Ladders	Safe entry and exit by entrants.	Daily inspection, inspection per manufacturer's recommendations.



Lighting equipment	Employees to see well enough to work safely and to exit the space quickly in an emergency.	Inspection per manufacturer's recommendations.
Personal protective equipment (PPE)	Protecting employees from hazards of the space or the work performed. CS PPE includes helmets, harnesses, gloves and boots.	Daily inspection, inspection per manufacturer's recommendations.
Rescue and emergency equipment, except for equipment provided by the rescue service provider	Safe and effective rescue. Skeds are used for packaging patients for horizontal drags.	Daily inspection, inspection per manufacturer's recommendations.
Testing and monitoring equipment.	Evaluating permit required confined space conditions by remote monitoring and attachment to entrant. Docking stations perform bump testing, calibration and documentation automatically.	Daily bump testing, calibration every 30 days.
Ventilating equipment	Obtaining and maintaining acceptable entry conditions.	Inspection per manufacturer's recommendations.
Any other equipment	Safe entry into and rescue from permit required confined spaces.	Before every use.

8. Record Keeping, Forms, and Program Review

8.1. Program Review

At a minimum, this Confined Space program will be reviewed every 3 years when there is any reason to believe that employees are not adequately protected and revised as necessary.

Situations that require this review include:

- Unauthorized entry of a permit space.
- Discovery of a previously unrecognized hazard.
- Existence of a condition prohibited by the permit or permit program.
- An injury or near-miss during entry.
- An employee reports concerns about the effectiveness of the program.
- Any other condition that affects employee safety or health.
- The request of a Workplace Safety Committee.



When revising the permit program to correct hazard related deficiencies, do not allow entries into affected permit spaces to be made until the revisions are complete.

Employees and their representatives can access the revised Confined Space program on the TriNet safety page.

The Safety Department shall perform an annual review of permits to be used to determine if a Confined Space program review should be initiated in less than the 3 year review cycle.

8.2. Forms

This Standard requires the following forms, and each are considered part of the adopted program. Forms may be updated as necessary with approval from the SMS Manager.

All program documents and forms may be accessed on TriNet.

FORM NUMBER	TITLE	RESPONSIBLE PARTY	DOCUMENT RETENTION
SAF-8251-PER-CSEvalPermit	Confined Space Evaluation and Permit	CS Supervisor	Emailed to: ConfinedSpace@trimet.org 1 year
SAF-8252-PLAN-CSRescue	Confined Space Rescue Plan	CS Supervisor	1 year
SAF-8253-F-CSLog	Confined Space Log	Site Manager	Emailed to: ConfinedSpace@trimet.org Indefinite
SAF-8254-TG-CSA	Confined Space Awareness Training Guide	Confined Space Program Administrator	1 year
SAF-8255-TG-SAER	Confined Space Supervisor, Attendant, Entrant and Non-IDLH Entry Training Guide	Confined Space Program Administrator	3 years
SAF-8256-TG-SAER-R	Confined Space Supervisor, Attendant, Entrant and Non-IDLH Entry Training Guide	Confined Space Program Administrator	3 years

9. Definitions

TERM	DEFINITION
ACGIH	American Conference of Governmental Industrial Hygienists. This organization produces Threshold Limit Values for airborne contaminants. TriMet will use either the OAR Permissible Exposure Limits (PEL's), or the TLV's, whichever is lower.



Acceptable Entry Conditions	The conditions that must exist in a permit required confined space to allow safe entry and work.
Alternate Entry	A procedure to enter confined spaces if the only remaining hazard is atmospheric (toxic, oxygen deficiency or enriched, or flammable) and it is controlled by ventilation.
Attendant	An individual stationed outside one or more permit spaces to monitor the authorized entrants and who performs all attendants' duties assigned in the employer's permit space program.
Authorized Entrant	A person who is approved or assigned by the department head in charge of the entry to perform a specific type of duty or duties or to be at a specific location at the job site.
Blanking or Blinding	The absolute closure of a pipe, line, or duct by the fastening of a solid plate (such as a spectacle blind or a skillet blind) that completely covers the bore and that can withstand the maximum pressure of the pipe, line, or duct with no leakage beyond the plate.
Bonding	The joining of two or more items with an electrical conductor so that all ends joined have the same electrical charge or potential.
Buddy System	An arrangement in which persons are paired, as for mutual safety or assistance.
Calibration	The checking of a direct-reading instrument against an accurate standard (such as a calibration gas) to determine any deviation and correct for errors. <i>NOTE: A similar process may also be referred to as a "bump test" in which an instrument is tested with an accurate standard to ensure it is still reading correctly. For the purposes of this rule, a "bump test" performed in accordance with the manufacturer's instructions can be used to verify calibration.</i>
Confined Space	A space that meets all the following: - Large enough and so configured that an employee can fully enter the space and perform work. - Has limited or restricted means for entry and/or exit. - Is not designed for continuous human occupancy.
Confined Space Evaluation	Document used to determine if a confined space can be entered using Permit Required or Alternate Procedures.
Confined Space Supervisor – Competent Person	The person responsible for determining if acceptable entry conditions are present in a confined space where entry is planned, for authorizing entry and overseeing entry operations and for terminating entry as required. The person is competent in the calibration and use of portable gas testing equipment and has successfully demonstrated use of the equipment in the field and is able to recognize potential hazards inherent to confined space entry.
Continuous System	A Confined Space that meets all the following: Part of, and contiguous with, a larger confined space (for example, storm sewers, sanitary sewers, or steam tunnels)



	Subject to a potential release from the larger confined space that can overwhelm control measures and/or personal protective equipment, resulting in a hazard that is immediately dangerous to life and health.
Controlling Contractor	The employer that has overall responsibility for construction at a worksite. <i>Note: A controlling contractor who owns or manages a property is both a controlling contractor and a host employer.</i>
Double Block and Bleed	The closure of a line, duct, or pipe by closing and locking or tagging two in-line valves and by opening and locking or tagging a drain or vent valve in the line between the two closed valves.
Emergency	Any occurrence (including any failure of hazard control or monitoring equipment) or event internal or external to the permit space that could endanger entrants.
Entry (breaking the plane)	The action by which any part of an employee's body breaks the plane of an opening into a confined space. Entry (or entry operations) also refers to the period during which an employee occupies a confined space.
Entry Attendant	An individual stationed outside one or more permit required confined space(s) who monitors the authorized entrants and who performs all assigned attendant duties.
Entry Permit	Written authorization from the employer, controlling contractor, or host employer to enter a permit required confined space and perform work.
Entry Supervisor	The trained and certified person (such as the employer, foreman, or crew chief, or any other designated employee) responsible for determining if acceptable entry conditions are present at a permit space where entry is planned, and authorizing entry and overseeing entry operations, and terminating entry as required. <i>NOTE: An entry supervisor also may serve as an attendant or as an authorized entrant if that person is trained and equipped as required by this program for each role they fill.</i>
Hazardous Atmosphere	An atmosphere that may expose employees to the risk of death, incapacitation, and impairment of ability to self-rescue (that is, escape unaided from a permit space), injury, or acute illness from one or more of the following causes: Flammable gas, vapor, or mist in excess of 10% of its lower flammable limit (LFL). Airborne combustible dust at a concentration that meets or exceeds its LFL. <i>NOTE: This concentration may be approximated as a condition in which the dust obscures vision at a distance of five (5) feet or less.</i> <i>Atmospheric oxygen concentration below 19.5% or above 23.5%.</i> <i>An airborne concentration of 50% of OAR permissible exposure limit (PEL's) or ACGIH TLV's.</i>
Hazard Control	The action taken to reduce the level of any hazard inside a confined space using engineering methods (i.e. by isolation or ventilation) and then using these methods to maintain the reduced hazard level. Hazard control also refers to the



	engineering methods used for this purpose. Personal protective equipment is not a hazard control.
Host Employer	An employer who owns or manages the property on which confined space work is taking place.
Hot Work	Any work involving burning, welding or similar fire-producing operations. Also, any work that produces a source of ignition, such as grinding, drilling, or heating.
Hot Work Permit	The employer's written authorization to perform operations (for example, riveting, welding, cutting, burning, and heating) capable of providing a source of ignition.
Immediately Dangerous to Life or Health (IDLH)	Any condition that poses an immediate or delayed threat to life or that would cause irreversible adverse health effects or that would interfere with an individual's ability to escape unaided from a permit space.
Inerting	The displacement of the atmosphere in a permit space by a noncombustible gas (such as nitrogen) to such an extent that the resulting atmosphere is noncombustible. <i>NOTE: This procedure produces an IDLH oxygen-deficient atmosphere.</i>
Lower Explosive Limit (LEL)	The minimum concentration of a combustible gas or vapor in air that will ignite if an ignition source is introduced.
Mobile Worker	An employee who performs work in multiple locations such as customer sites, company offices, private homes, vendor offices, or construction sites.
Non-Entry Rescue	Retrieval of entrants from a permit space without entering the permit space.
Permit Required Confined Space (Permit Space)	A confined space that has one or more of the following characteristics: Contains, or has a potential to contain, a hazardous atmosphere. Contains a material that has the potential to engulf an entrant. Has an internal configuration such that an entrant could become trapped or asphyxiated by inwardly converging walls or by a floor which slopes downward and tapers to a smaller cross-section. Contains any other recognized serious safety or health hazard that can inhibit an entrant's ability to escape unaided.
Physical Hazard	An existing or potential hazard that can cause death or serious physical harm in or near a confined space, or a hazard that has a reasonable probability of occurring in or near a confined space, and includes, but is not limited to: Explosives, mechanical, electrical, hydraulic, and pneumatic energy, radiation, temperature extremes, engulfment, noise, and inwardly converging surfaces, and chemicals that can cause death or serious physical harm through skin or eye contact (rather than through inhalation).
Potential Hazards	All reasonably anticipated conditions within the space and outside the space that can adversely affect conditions within the space.
Rescue	Retrieving employees who are unable to remove themselves from a permit space. Rescue can be entry or non-entry and can be conducted by the employer's employees or a third-party.

Retrieval System	The equipment, including mechanical retrieval devices, used for non-entry rescue of authorized entrants from a permit space.
Testing	The process of identifying and evaluating the atmospheric hazards that entrants may be exposed to in a permit required confined space. Testing includes specifying the initial tests that are to be performed in the permit space (see also “monitor” or “monitoring”).
Ventilate or Ventilation	Controlling an actual or potentially hazardous atmosphere using either powered equipment, such as fans and blowers, or reliable natural air flow, or a combination of the two, to reduce an otherwise hazardous atmosphere below the level that makes it a hazardous atmosphere. Ventilation is a method of hazard control, not hazard elimination.

10. References

REFERENCE	CITATION	COMMENTS / LINKS
OAR	437-002-0146	<u>Confined Spaces</u>
OSHA	1910.146	<u>Permit Required Confined Spaces</u>
OSHA	1926 Subpart AA	<u>Confined Space in Construction</u>

11. Version History

VERSION	DATE	SUMMARY OF CHANGES	OWNER	APPROVED BY
1.0	03/06/25	Original Program	SMS Manager Jessie Harvey	